

i Examination paper for IT3708 – Bio-inspired Artificial Intelligence

Date: May 13, 2025

Time: 09:00

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Present at the exam location: NO

Permitted examination support material: D / No printed or hand-written support material is allowed. A specific basic calculator is allowed.

OTHER INFORMATION

Read the questions carefully and make your own assumptions. In your answers, explain clearly what assumptions you have made and how you have understood or limited the assignment.

If there are direct errors or omissions in the assignment set and you cannot make your own assumptions, please refer to the information about complaints regarding formal errors on the NTNU website “Explanation of grades and appeals”.

SPECIFIC INFORMATION FOR YOUR COURSE

No paper drawings:

This exam does not include hand drawings. If you receive hand drawing sheets, this is by mistake. You will not be able to submit the sheets, and they will not be graded.

Weighting:

Each “multiple choice” question has exactly one answer and is worth 1 point, for a total of 16 points. The “written answer” questions are worth a total of 14 points. For each “written answer” question, the maximum score is given with the question. The total number of points for the exam is 30.

Withdrawing from the exam:

If you wish to submit a blank test/withdraw from the exam for another reason, go to the menu in the top right-hand corner and click “Submit blank”. This cannot be undone, even if the test is still open.

Access to your answers:

After the exam, you can find your answers under previous tests in Inspera. Be aware that it may take a working day until any hand-written material is available in “previous tests”.

- 1 Among the following statements, pick one **correct statement** about the role of diversity maintenance in multi-objective optimization using genetic algorithms.

Select one alternative:

- ☐ Once a set of non-dominated solutions is found, diversity maintenance can be discontinued, as the algorithm will automatically preserve a well-distributed Pareto front throughout subsequent generations.
- ☐ Maintaining diversity in multi-objective optimization is achieved most effectively by repeatedly selecting identical individuals from the population to reinforce convergence toward a single optimal solution.
- ☐ Diversity mechanisms in multi-objective genetic algorithms are only applicable in single-objective problems and are considered counterproductive when handling more than one objective function.
- ☐ Diversity maintenance in multi-objective genetic algorithms is crucial for ensuring a well-distributed Pareto front, enabling the algorithm to explore trade-offs effectively and avoid premature convergence to suboptimal regions. ✓
- ☐ Diversity maintenance in multi-objective genetic algorithms is unnecessary, as the algorithm naturally produces evenly distributed solutions across the entire Pareto front without any specialized techniques.

Maximum marks: 1

2 Among the following statements, pick one **correct statement** about the role of multi-objective optimization in bio-inspired algorithms.

Select one alternative:

- ☒ Multi-objective optimization plays a pivotal role in bio-inspired algorithms by enabling the simultaneous exploration of trade-offs among conflicting objectives, often yielding a diverse set of Pareto-optimal solutions. ✓
- ☐ Bio-inspired algorithms do not require multi-objective optimization techniques because natural systems inherently evolve toward only one optimal solution in all environmental contexts.
- ☐ All bio-inspired algorithms that apply multi-objective optimization techniques are guaranteed to converge to the true Pareto front within a fixed number of generations, regardless of problem complexity.
- ☐ The role of multi-objective optimization in bio-inspired algorithms centers on decorative visualizations of trade-off curves and holds no practical significance in real-world engineering applications.
- ☐ In bio-inspired algorithms, multi-objective optimization operates under the assumption that all objectives are linearly dependent, thereby simplifying the problem into a single-objective model.

Maximum marks: 1

- 3 Among the following statements, pick one **correct statement** about the role of mutation and crossover in multi-objective genetic algorithms.

Select one alternative:

- ☐ In multi-objective genetic algorithms, the crossover operator is primarily responsible for evaluating objective functions and does not contribute to the generation of new candidate solutions.
- ☐ In multi-objective genetic algorithms, mutation and crossover are fundamental variation operators that generate diverse offspring and facilitate exploration of the solution space ✓ approximate the Pareto-optimal front.
- ☐ The use of both crossover and mutation in multi-objective genetic algorithms is discouraged, as the operators disrupt the non-dominated sorting process and introduce inconsistencies in Pareto front estimation.
- ☐ In multi-objective genetic algorithms, the mutation operator acts as a selection mechanism, filtering out infeasible individuals rather than introducing variation into the population.
- ☐ Mutation in the NSGA-II multi-objective genetic algorithm is deterministic and designed to move solutions directly toward the center of the objective space, ignoring diversity preservation.

Maximum marks: 1

- 4 Among the following statements, pick one **correct statement**. The statements are concerned about the use of Markov chains to analyze genetic algorithms.

Select one alternative:

- ☐ Due to the fact that genetic algorithms are designed primarily for empirical applications, formal analysis through Markov chains is impossible and offers no theoretical insights.
- ☐ The ease of using Markov chains to analyze genetic algorithms derives from the assumption that populations consist of independent individuals, which eliminates interactions and simplifies modeling efforts.
- ☐ Because genetic algorithms operate deterministically once initialized, Markov chain analysis can reliably predict exact future populations without consideration of random variation introduced by crossover or mutation.
- ☐ Analyzing genetic algorithms by means of Markov chains is in principle possible but in practice difficult, due to large state spaces and stochastic transitions. ✓
- ☐ Genetic algorithms inherently induce small and manageable state spaces, enabling straightforward construction of transition matrices and exact calculation of steady-state distributions via standard Markov chain modeling.

Maximum marks: 1

- 5 Among the following statements about the role of computational complexity theory in bio-inspired algorithms, which one is **correct**?

Select one alternative:

- ☐ Computational complexity theory is not relevant to bio-inspired algorithms, because their stochastic nature allows them to solve any problem, including undecidable ones, without concern for time or resource constraints.
- ☐ Complexity theory discourages the use of bio-inspired algorithms, as it classifies all heuristic and metaheuristic methods as inefficient and fundamentally unprovable in terms of performance or scalability.
- ☐ Computational complexity theory helps in understanding the limitations of bio-inspired algorithms, especially for NP-hard problems, where they offer approximate or heuristic solutions without guarantees of optimality or polynomial runtime. ✓
- ☐ Bio-inspired algorithms are not affected by NP-hardness since they use randomization, which transforms any hard problem into an easy one through iteration over a number of generations with an arbitrarily sized population, thus mimicking natural evolution.
- ☐ There is no connection between computational complexity theory and bio-inspired algorithms, as the former deals only with theoretical models while the latter operates entirely outside of formal analysis including computational complexity theory.

Maximum marks: 1

6 What are the pros and cons of using genetic algorithms to solve NP-complete problems? Among the following statements, which statement is **correct**?

Select one alternative:

- ☐ Since genetic algorithms can be formalized by means of Markov chains informed by the algorithms' operators and parameters, these algorithms are not impacted by the NP-hardness results of computational complexity theory.
- ☐ As genetic algorithms mimic natural evolution, they typically avoid local minima and guarantee finding global optima in problems of practical interest, regardless of their computational complexity or dimensionality.
- ☐ The performance of a genetic algorithm is strongly affected by the encoding of the problem it is solving, and as a consequence computational complexity depends on the algorithm's encoding choice, rather than the nature of the problem itself.
- ☐ Genetic algorithms provide a useful heuristic approach for approximating solutions to NP-complete problems, though they don't guarantee optimality and can be sensitive to their parameters and problem encoding. ✓
- ☐ Using genetic algorithms is always more efficient than traditional heuristics, as they require minimal parameter tuning and perform close to optimally with default crossover and mutation rates, making them suitable to any NP-complete problem.

Maximum marks: 1

- 7 The concept of neighborhood is crucial to define a search landscape. Which of the following statements about the neighborhood is **false**?

Select one alternative:

- ☐ The neighborhood of a solution x is the set of solutions that are within a given distance of x .
- ☒ Two bitstrings have the same neighborhood if they have the same number of 1s. ✓
- ☐ The neighborhood of a bitstring b is usually considered as the set of bitstrings that can be reached with one bitflip.
- ☐ The neighborhood in a permutation vector is, for example, the set of all permutations that can be reached in one mutation step.
- ☐ A bitstring of size n has up to n neighbors when we define the neighborhood as having a Hamming distance of size 1.

Maximum marks: 1

- 8 Particle Swarm Optimization (PSO) is a metaheuristic inspired by bird flocks. Which of the following statements about PSO is **false**?

Select one alternative:

- ☐ Particles in PSO consider the positions and velocities of their neighbors.
- ☐ Each particle in PSO adds pheromone to a trail, making it more likely to be followed by other particles in the swarm.
- ☐ In PSO, the size of a particle's neighborhood will define how many particles will influence that particle's movement.
- ☐ PSO was proposed as a metaheuristic for continuous optimization.
- ☐ In PSO, each particle has a position, a velocity, and keeps a record of its personal best position.

Maximum marks: 1

- 9 In constrained optimization, the final solution needs to be feasible. Which of the following statements is **true** about feasibility?

Select one alternative:

- ☐ Performing mutation on an infeasible solution will ensure feasibility if the mutation probability parameter is high enough.
- ☐ A feasible solution will always have a better fitness than an infeasible solution since the infeasible solution will be heavily penalized.
- ☐ Performing crossover of two feasible solutions will ensure that the offspring are feasible solutions.
- ☐ Searching only in the feasible areas of a search landscape will not ensure reaching the global optimum. ✓
- ☐ Starting with a 100% feasible population will ensure that the population of the next generation is 100% feasible.

Maximum marks: 1

- 10 Throughout the course, we studied several problems; both discrete (or combinatorial) and continuous. Which of the following problems is **not** combinatorial?

Select one alternative:

- ☐ Vehicle Routing Problem with time windows
- ☐ The n -dimensional Ackley Function ✓
- ☐ Feature Selection
- ☐ Travelling Salesperson Problem
- ☐ Binary Knapsack Problem

Maximum marks: 1

11 Which of the following statements is **not** a way to escape local optima?

Select one alternative:

- ☒ Use elitism to increase the selective pressure
- ☐ Apply perturbations to promising solutions
- ☐ Gradually reduce the frequency of taking a non-optimal decision
- ☐ Make non-optimal decisions every so often
- ☐ Restart the search process



Maximum marks: 1

12 During the course, we studied search landscapes to understand how our algorithms traverse the search space. Which of the following statements about landscape analysis is **false**?

Select one alternative:

- ☐ Different algorithms will create different search landscapes of the same fitness function
- ☐ Fitness landscapes were originally proposed in the study of Genetics (to explain why some species stopped evolving)
- ☒ A search landscape can be generated by running an algorithm until it finds all local optima in the search space
- ☐ Studying search landscapes can help assessing problem difficulty and explain algorithm behavior
- ☐ A search landscape depends on a fitness function, a search space, and a neighborhood definition



Maximum marks: 1

- 13 As in solvers in machine learning, many of the metaheuristics and algorithms in evolutionary computation need some sort of parameter setting. Which of the following statements regarding parameters is **false**?

Select one alternative:

- ☐ Parameter tuning is performed by the user, and it relies on the user's knowledge about the metaheuristic and about the problem being solved.
- ☐ Setting a rule for increasing the mutation rate every k generations is a form of deterministic parameter control.
- ☐ Encoding mutation rates as part of an individual's chromosome is a form of self-adaptive parameter control.
- ☐ Parameters can be set up and adjusted before the run and during the run, which we call parameter tuning and parameter control, respectively.
- ☐ Increasing the mutation rate after every k successful mutations in Evolutionary Strategy is a form of self-adaptive parameter control. ✓

Maximum marks: 1

- 14 Which of the following statements is **not true** about the single point crossover (1PX) when applied in constrained optimization?

Select one alternative:

- ☐ By using 1PX, genes that are near each other are more likely to be kept together.
- ☐ 1PX is an exploitation technique.
- ☐ 1PX works for both continuous and binary representations in constrained optimization.
- ☐ Using 1PX on two feasible individuals (when dealing with permutation vectors) ensures that the crossover results in at least one feasible solution. ✓
- ☐ 1PX can never keep together genes from opposite ends of the string.

Maximum marks: 1

- 15 Evaluate the following statements about the population management in an evolutionary algorithm and pick the one that is **true**.

Select one alternative:

- ☒ Selection from the population does not depend on the chromosome representation. ✓
- ☐ Selection in evolutionary algorithms is often performed by means of local search.
- ☐ How selection and replacement work together does not need to be considered.
- ☐ When you perform selection in a genetic algorithm, there is no need to do replacement.
- ☐ The choice of a selection method depends entirely on the crossover method used.

Maximum marks: 1

- 16 Memetic Algorithms (MA) are hybrid algorithms that integrate an Evolutionary Algorithm and some other algorithm, for example Stochastic Local Search. Which of the following statements is **not** true about MAs?

Select one alternative:

- ☐ When including known good solutions, average performance is often increased. However, best performance can come from a randomized inclusion of solutions
- ☐ Constructive heuristics are commonly used, for example "Maximum number of conflicts first" in Constraint Optimization Problems
- ☐ To incorporate good solutions into the population, Stochastic Local Search can be used to climb from random individuals in the population
- ☐ Initializing populations with already known solutions is a common practice in MAs
- ☐ Since Local Search considers neighborhoods, it can only be applied to permutation problems ✓

Maximum marks: 1

Answer 17

The first way of speeding up the search in an EA is during the design phase. Choosing a good **individual representation** helps **reducing the size of the search space**. Then, during implementation using the **appropriate data structures** (for example matrices and arrays instead of dictionaries) can bring extra speed benefits in **scientific programming languages** that are compiled (like C/C++, Julia, or Rust).

Another way of speeding up an EA is by **keeping the number of fitness evaluations to a minimum**, for example, by using a **hash map** to **avoid redundant computations**. In a similar fashion, a **surrogate model** can be used to **approximate fitness functions** that are very expensive, for example, by training a machine learning model on a dataset of a sample of fitness evaluations.

Finally, using **parallelization** (either via GPUs, multithreading, multiprocessing or message passing) can greatly increase the performance of an EA. Other **code optimization** tricks like type declarations, vectorization and broadcasting, can also be useful to help the compiler **avoid unnecessary computations**.

Answer 18

Answer 1: Even though the problem is computationally hard, the client may have been insisting on the development of a polynomial-time algorithm. The recent discovery of the proof that the problem is NP-complete, along with the assumptions that (1) P is not equal to NP and (2) the NP-completeness result for the problem is correct, brings new light to this issue. Either the client needs to give up on the search for a polynomial-time algorithm to solve the problem (but see Answer 2 and Answer 3) or they need to give up on solving the problem at all.

Answer 2: While the original problem is NP-complete, it may be possible to restrict that problem. This means focusing on a subset of the problem instances and develop a polynomial-time algorithm for that subset. Even if the polynomial-time algorithm cannot handle all instances of the original problem, the client may be satisfied. For example, it may be the case that only such a subset of problem instances shows up in the client's software application.

Answer 3: It is an option to develop an algorithm that seeks to handle all problem instances of the problem, knowing that there will be limits to the algorithm. A heuristic bio-inspired algorithm such as a genetic algorithm can be a reasonable method to approximately solve the client's NP-complete problem relatively fast, but there will not be a guarantee of finding a solution even when it exists. Since solutions can be verified in polynomial time, this can be utilized as a termination condition for the heuristic algorithm. Overall, the approach will then follow a generate-and-test pattern where the generate part is the heuristic algorithm while the test part will be the polynomial-time verification algorithm.

Answer 19

Experimental approach: Experimentally, one would repeatedly run an implementation of the bio-inspired algorithm A under similar conditions for both the PeakedOneMax and OneMax problems. To produce statistically significant results, we would record the performance of the implementation in optimizing OneMax and PeakedOnemax, for example in terms of the number of fitness evaluations needed until the global optimum is found. A global optimum is considered found when A encounters it in the population. The experimental results should show that PeakedOneMax is harder to solve than normal OneMax, due to the deceptive local but non-global optimum at all-1s (or 11...1) for the PeakedOneMax problem, while this is the true global optimum for the OneMax problem.

Analytical approach: The problems PeakedOneMax and OneMax and the bio-inspired algorithm A can be analyzed using discrete-time, discrete-space Markov chains in which the states represent possible populations of A and the Markov chain probabilities reflect A's operations. In this way, one can create Markov chains that formalize how algorithm A ends up with populations containing globally optimal solutions for PeakedOneMax and OneMax, respectively. By comparing the expected hitting times of the Markov chains for the two problems, we will observe a longer expected hitting time for PeakedOneMax compared to that of OneMax, reflecting the former problem's deceptive optimum at all-1s (or 11...1).